

# Indian Census 2011 Educational Data

## Educational Attainment by Age and Sex — Project Report

### 1. Executive Summary

This project analyses Census 2011 educational attainment using the supplied C-08 data. It combines a documented Pandas cleaning notebook, cleaned datasets, SQL analysis, a React/TypeScript dashboard preview and a Power BI implementation guide. The central methodological issue is the literacy denominator: the supplied C-08 file includes a special 0–6 row treated as illiterate. Subtracting it from the All ages total gives a 7+ denominator of 1,046,339,724 and a directly derived literacy rate of 72.98%. The widely published Census 2011 headline rate is 74.04%, so the project should distinguish the C-08-derived rate from the headline statistic.

### 2. Objectives

Clean the C-08 table; create a long-format fact table; analyse literacy by sex and age; compare rural and urban outcomes; identify State/UT differences; and produce reproducible SQL and dashboard outputs.

### 3. Source and Coverage

The official Census metadata describes C-08 as educational level by age and sex for population age 7 and above, with Total/Rural/Urban residence and single-year ages 7–19 followed by grouped ages. The supplied data contain 36 geographic units, 3 residence categories, 29 age groups, 13 education categories and 122,148 long-format rows.

### 4. Data Cleaning

The notebook reconstructs the merged header, removes title/header/artifact rows, standardises age labels, cleans geography names, removes constant metadata and the empty trailing column, converts counts to nullable integers, and runs sanity checks. It expects the original DDW-0000C-08.xlsx workbook, which is not included in the supplied project bundle.

### 5. Analytical Definitions

7+ Population = All-age Population – 0–6 Population. C-08 Derived Literacy Rate = Literate Population ÷ 7+ Population × 100. Male–Female Gap = Male Literacy Rate – Female Literacy Rate. Urban–Rural Gap = Urban Literacy Rate – Rural Literacy Rate. Graduate Share = Graduate & Above ÷ Literate × 100.

### 6. National Results

| Indicator             | Result        |
|-----------------------|---------------|
| Total population      | 1,210,854,977 |
| 0–6 population        | 164,515,253   |
| 7+ population         | 1,046,339,724 |
| Literate              | 763,638,812   |
| Illiterate            | 447,216,165   |
| C-08-derived literacy | 72.98%        |
| Male literacy         | 80.88%        |
| Female literacy       | 64.63%        |
| Male–female gap       | 16.25 pp      |
| Rural literacy        | 67.77%        |
| Urban literacy        | 84.11%        |
| Rural–urban gap       | 16.34 pp      |
| Graduate & above      | 68,288,971    |

## 7. State/UT Findings

| Rank | Highest     | Rate   |
|------|-------------|--------|
| 1    | Kerala      | 94.00% |
| 2    | Lakshadweep | 91.85% |
| 3    | Mizoram     | 91.33% |
| 4    | Goa         | 88.70% |
| 5    | Tripura     | 87.22% |

| Rank | Lowest            | Rate   |
|------|-------------------|--------|
| 1    | Bihar             | 61.80% |
| 2    | Arunachal Pradesh | 65.38% |
| 3    | Rajasthan         | 66.11% |
| 4    | Jharkhand         | 66.41% |
| 5    | Andhra Pradesh    | 67.02% |

## 8. Sex/Gender Gap

| State/UT             | Male   | Female | Gap      |
|----------------------|--------|--------|----------|
| Rajasthan            | 79.19% | 52.12% | 27.07 pp |
| Jharkhand            | 76.84% | 55.42% | 21.42 pp |
| Dadra & Nagar Haveli | 85.17% | 64.32% | 20.86 pp |
| Jammu & Kashmir      | 76.75% | 56.43% | 20.32 pp |
| Uttar Pradesh        | 77.28% | 57.18% | 20.10 pp |

## 9. Rural–Urban Gap

| State/UT             | Rural  | Urban  | Gap      |
|----------------------|--------|--------|----------|
| Dadra & Nagar Haveli | 64.12% | 89.79% | 25.66 pp |
| Arunachal Pradesh    | 59.94% | 82.93% | 22.99 pp |
| Jharkhand            | 61.11% | 82.26% | 21.14 pp |
| Meghalaya            | 69.92% | 90.79% | 20.87 pp |
| Andhra Pradesh       | 60.45% | 80.09% | 19.64 pp |

## 10. Age Pattern

| Age   | Illiteracy | Graduate & Above |
|-------|------------|------------------|
| 7     | 27.18%     | 0.00%            |
| 10    | 11.31%     | 0.00%            |
| 15    | 10.04%     | 0.00%            |
| 20–24 | 16.76%     | 11.82%           |
| 35–39 | 32.73%     | 9.60%            |
| 50–54 | 42.43%     | 7.53%            |
| 65–69 | 56.06%     | 4.12%            |
| 80+   | 61.04%     | 3.01%            |

This should be interpreted as a descriptive cross-sectional cohort pattern in 2011, not as a causal effect of age.

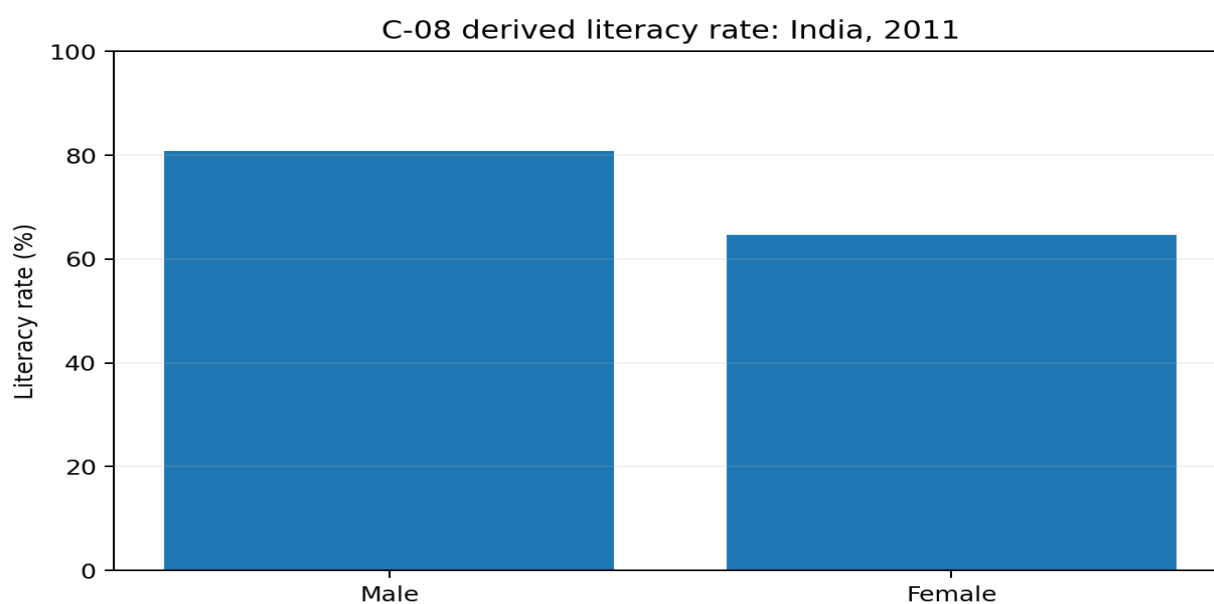
## 11. Educational Attainment Mix

| Category                           | Share of literates |
|------------------------------------|--------------------|
| Primary                            | 24.12%             |
| Below Primary                      | 19.24%             |
| Middle                             | 17.53%             |
| Matric/Secondary                   | 13.87%             |
| Higher Secondary                   | 10.21%             |
| Graduate & Above                   | 8.94%              |
| Literate without educational level | 4.60%              |
| Technical Diploma                  | 0.95%              |
| Unclassified                       | 0.40%              |
| Non-Technical Diploma              | 0.14%              |

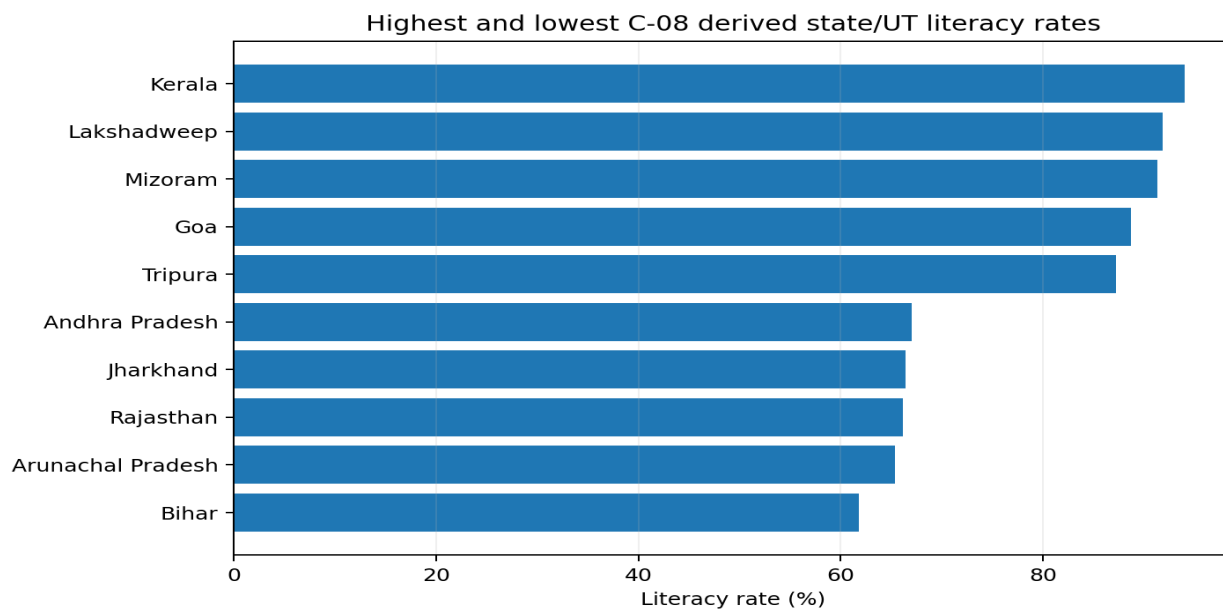
## 12. SQL and Dashboard Review

The SQL layer contains ten coherent analytical questions. Q1, Q2, Q3, Q6 and Q10 should be revised if an effective 7+ denominator is intended, because the current versions use all-age totals in relevant places. The dashboard contains five pages: Overview, Gender Gap, Rural vs Urban, Age & Attainment and Education Mix. The Power BI guide recommends separating aggregate levels from detailed attainment categories to avoid double-counting.

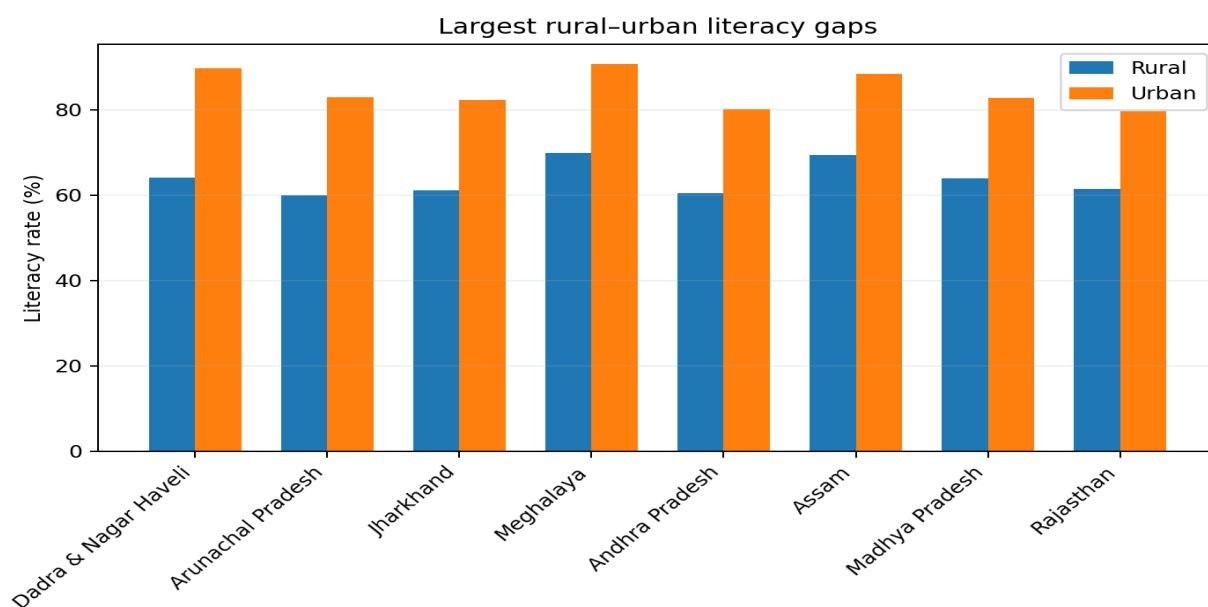
## 13. Visual Evidence



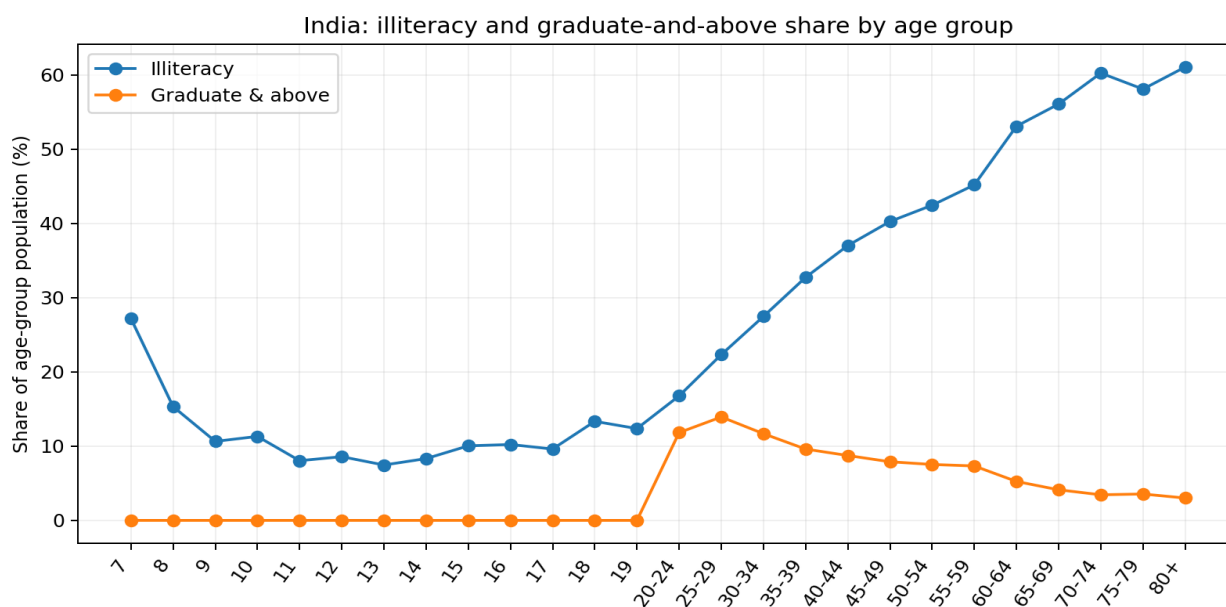
National C-08-derived male and female literacy.



Highest and lowest State/UT C-08-derived literacy.



Largest rural-urban literacy gaps.



## 14. Limitations

- 2011 data are historical, not current estimates.
- C-08 measures literacy/educational level, not current school attendance.
- Sex categories are limited to Male/Female.
- Age is single-year only for 7–19 and grouped thereafter.
- State/UT geography follows the 2011 structure.
- The C-08-derived rate differs from the widely published 74.04% headline rate.
- The raw DDW-0000C-08.xlsx workbook is not included in the supplied project bundle.
- Descriptive differences should not be interpreted as causal effects.

## 15. Recommendations

- Standardise the 7+ denominator across Python, SQL, Power BI and React.
- Separate aggregate and attainment categories.
- Add automated data-quality tests.
- Add a numeric age sort key.
- Add a methodology page/card.
- Add insight statements to dashboard pages.
- Add a data dictionary and screenshots to GitHub.
- Keep the denominator discrepancy visible and explained.

## 16. Making the Project Feel Authored and Credible

The project should not attempt to disguise AI assistance. Make the analytical decisions clearly your own: write a short motivation paragraph, add observations you personally noticed, execute the notebook, include screenshots, maintain genuine commit history, and be able to explain every SQL query and visual. Do not fabricate personal experiences or claim the project is AI-free if it is not.

Suggested disclosure: “Generative AI was used selectively for documentation, formatting and code-review support. The underlying Census files, calculations and final analytical decisions were reviewed against the project data and source documentation.”

## 17. Final Assessment

The technical foundation is strong. The main improvement is methodological consistency: use one denominator convention across Python, SQL, Power BI and React and distinguish the C-08-derived 72.98% rate from the standard published Census 2011 headline rate.

## 18. Source

Office of the Registrar General & Census Commissioner, India. Population Census 2011. C-08: Educational level by age and sex for population age 7 and above (total), India.

<https://censusindia.gov.in/nada/index.php/catalog/44790>